Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	README later
Selected Domain	Domain 2: Orbyte Smart SME Challenge: Less Busywork, Better Business
Selected SDG Track (1–17)	SDG 8: Decent Work and Economic Growth
Project / Solution Name	Halal Supplier Control
Name (Leader)	Hu Jay
Email & Phone (Leader)	hu.jay.hu.jay@gmail.com & +6012-942 8971
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

A halal-certified micro F&B business in Malaysia must prove at any audit that every ingredient comes from a supplier with a current, in-scope halal certificate. Under MHMS 2020, certified micro and small premises run an Internal Halal Control System (IHCS), and its core record is the Raw Material Masterlist: each material's supplier, manufacturer, certificate and expiry. However, in a micro-SME that record is kept in folders, WhatsApp photos and spreadsheets, kept by an owner or Halal Supervisor who is also running the kitchen.

Two failures follow. Supplier certificates expire unnoticed, and an expired certificate counts the same as none. Suppliers also deliver a different brand or manufacturer of the same ingredient, and nobody notices until an auditor compares the labels with the masterlist. A failed audit can lead to suspension of the certificate and the loss of customers who depend on it.

The work is also repeated across the industry. Every buyer chases every supplier for the same certificate, so ten bakeries buying from one flour supplier keep ten copies that go stale separately. Existing tools don't cover delivery-time control for a micro SME: consumer halal scanners serve shoppers, Verify Halal is a manual lookup, and enterprise halal systems target manufacturers and warehouses.

The segment is large: Malaysian MSMEs produced 39.7% of GDP and 48.7% of employment in 2025 (DOSM, 2026).

Source: Department of Statistics Malaysia (DOSM), "Micro, Small & Medium Enterprises (MSMEs) Performance 2025", 30 July 2026, https://www.dosm.gov.my/site/downloadrelease?id=micro-small--medium-enterprises-msmes-performance-2025&lang=English.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Halal Supplier Control is an Orbyte module that checks each delivered ingredient at the receiving door. Staff choose the material and photograph its label. A vision-language model, Qwen3-VL-8B-Instruct, reads the product name, brand, manufacturer and whether a halal mark is present, and returns them as structured JSON. AI is used only for this reading step. Five Python rules then decide the verdict by checking that the reading is complete, that the brand and manufacturer match the masterlist, that the supplier's certificate covers the product and hasn't expired, and that the issuer is recognised.

A Covered delivery is logged automatically. Every other verdict (Substitution, Not covered or Reading incomplete) waits in the Halal Supervisor's queue. The supervisor sees the photo, the fields read from it and the rule checklist, then accepts or rejects with a required reason. Every upload, verdict and decision goes into an activity log with the role, action and time.

Supplier certificates are stored in a shared registry. When a supplier renews once, every SME that buys from that supplier sees its own items change from Expiring to Covered. This works only because many SMEs and their suppliers share one platform. The system reports document coverage and gives no halal ruling.

From the SDC Phase 1 workshops, the prototype uses a GenAI API (the model served on RunPod through an OpenAI-compatible API, with a JSON schema fixing the output), database architecture (PostgreSQL with organisation-scoped queries and an audit log) and cloud hosting (a Replit Autoscale deployment with managed secrets). The prototype is live at https://halal-supplier-control--dedoh.replit.app, and the full journey, including the supplier renewal, runs end to end on it.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

- Frontend: React 19 with Vite and Tailwind CSS, built to static files.
- Backend: FastAPI (Python), which serves the built frontend and the JSON API.
- Database: PostgreSQL on Replit through SQLAlchemy.
- AI model: Qwen3-VL-8B-Instruct on a RunPod Serverless endpoint with the vLLM worker, called through the OpenAI Python SDK with a JSON schema for the output.
- Decision logic: five Python rules, with rapidfuzz for brand and manufacturer matching.
- Hosting: one Replit Autoscale deployment, with the RunPod key in Replit Secrets.

System block diagram (**Appendix A**): the browser talks to FastAPI on Replit, which reads and writes Replit PostgreSQL and calls the RunPod endpoint for label reading.

Sponsor resources: yes, both Replit and RunPod are used. Replit and RunPod both run the working product. Replit Autoscale hosts the whole app: FastAPI serves the React screens and the API, Replit PostgreSQL stores the supplier records, certificates, delivery photos, decisions and activity log, and Replit Secrets hold the RunPod key. RunPod Serverless runs Qwen3-VL-8B-Instruct on a 24 GB GPU to read each label photo. It scales to zero between checks, so cost follows use.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise and define: Our observations of halal-certified F&B operations highlighted the challenge of maintaining compliance alongside daily production and delivery receiving. We focused on two risks: expired supplier certificates and unnoticed brand substitutions.

Ideate: These observations shaped a workflow where AI reads ingredient labels, five explicit rules generate a recommendation, and the supervisor retains the final decision. This keeps outcomes explainable and accountable. A shared supplier registry reduces repeated certificate collection across buyers.

Prototype and test: We built a running MVP, evaluated label extraction on eight generated images covering clean, substituted and unreadable labels, and completed the seven-step journey on the deployed app. This demonstrates initial technical feasibility; broader accuracy and usability validation remains necessary.

Deploy and improve: The MVP runs on Replit Autoscale with PostgreSQL, with AI inference hosted on RunPod Serverless. Phase 2 will add certificate-photo prefilling, WhatsApp submission through the Meta Cloud API, certificate cross-checks against Verify Halal or MYeHALAL, and scheduled expiry monitoring with supplier reminders.

We will pilot the module with one halal-certified premise, compare checking times against our manual baseline, and evaluate a larger label dataset. Staff task completion and understanding of flagged discrepancies will guide refinements before wider deployment.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 8.2: "Achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including through a focus on high-value added and labour-intensive sectors." Target 8.3: "Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Checked by hand, one delivery took a median of 82 seconds (59 to 112) on our 8 synthetic deliveries. Today the check is fully manual: a person compares each label with the masterlist and the supplier's certificate. Deliveries decided without the certificate took 59 to 66 seconds, and deliveries that needed the certificate took 98 to 112 seconds. One team member timed each case once. We found no published figure to compare with.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Time to check one delivery, in seconds, from choosing the material to reading the verdict, reported as the percentage reduction against the 82-second manual median.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	At least 70% less time per delivery check than a manual check of the same labels, so about 25 seconds or less against today's 82-second median. Both are timed on a larger set of labels held out from prompt tuning. The time saving counts only while at least 90% of seeded substitutions and 100% of seeded expired or out-of-scope certificates are still flagged.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Qwen3-VL-8B-Instruct	Model	Apache-2.0
vLLM	Code (model server)	Apache-2.0
FastAPI, Uvicorn	Code	MIT; BSD-3-Clause
SQLAlchemy, psycopg	Code	MIT; LGPL-3.0
Pydantic, python-multipart	Code	MIT; Apache-2.0

OpenAI Python SDK	Code	Apache-2.0
rapidfuzz, Pillow	Code	MIT; MIT-CMU
React, Vite, Tailwind CSS, TanStack Query	Code	MIT
wouter, react-aria-components, lucide-react	Code	Unlicense; Apache-2.0; ISC
Rambla font	Font	OFL-1.1

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

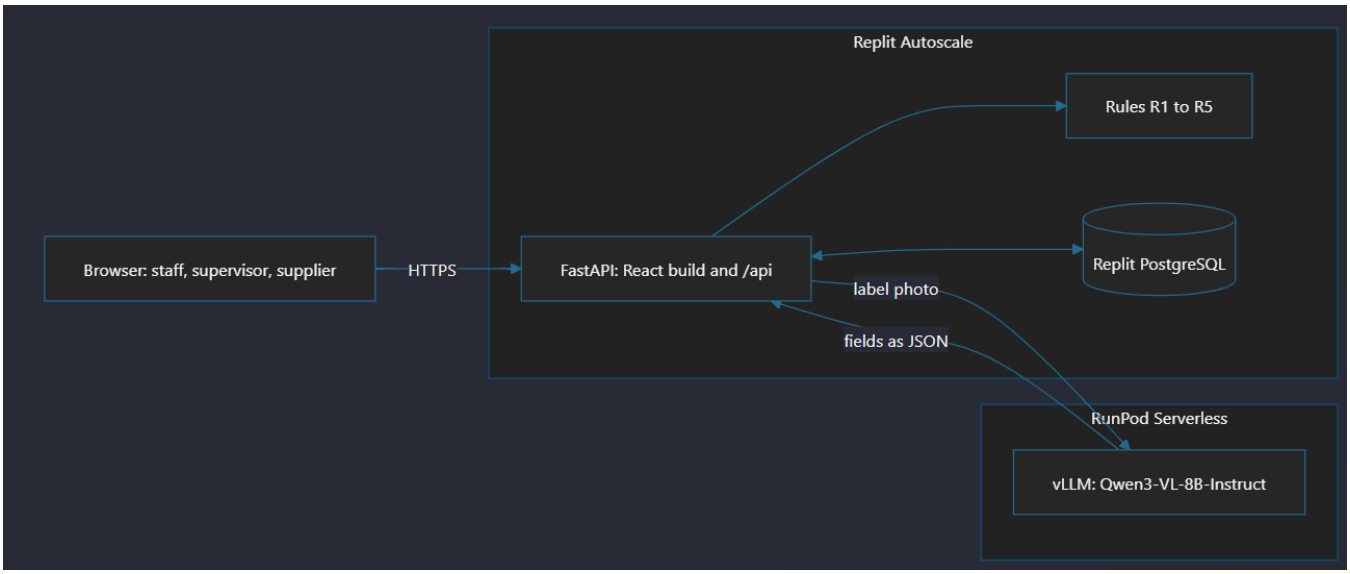
7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24/10/2026

Appendix A


Replit Autoscale
Rules R1 to R5
Browser: staff, supervisor, supplier
HTTPS
FastAPI: React build and /api
Replit PostgreSQL
label photo
fields as JSON
RunPod Serverless
vLLM: Qwen3-VL-8B-Instruct